SEMESTER S5

POWER SYSTEMS

Course Code	PCEOT503	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GYEST104	Course Type	Theory

Course Objectives:

1. To deliver fundamental concepts in power system components.
2. To deliver basic idea of power generation, transmission and protection.
3. To deliver fundamental concepts of protection in power system.
4. To deliver fundamental concepts of steady state and transient analysis in power system.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Generation from renewable and non-renewable sources – Hydro, thermal, nuclear- (block schematic details, environmental and ethical factors, advantages, disadvantages) Solar and wind - (block schematic details, environmental factors, regulations, advantages, disadvantages) Energy storage systems as alternative energy sources – BESS, CESS, thermal SS Load curve – Load duration curve, Load factor, diversity factor, demand factor, Plant capacity factor, plant use factor - Numerical Problems.	11
2	Power Transmission System - (Electrical Model) - Line parameters – resistance - inductance and capacitance (Derivation of three phase double circuit) Transmission line modelling - classifications (concept only) – transmission	11

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	line as two port network – derivation and calculation of ABCD parameters (derivation and numerical problems) Skin Effect & Ferranti Effect – Corona (qualitative study only) – Surge Impedance Loading Insulators – string efficiency – grading (numerical problems)	
3	Per unit quantities-single phase and three phase Symmetrical components - sequence networks Types of faults – Fault calculations(shunt only)-symmetrical and unsymmetrical Need for protection- Types of protection schemes – primary and back-up Protective relays – Basics of typical electromechanical relay – induction type only - Static (block diagrams of over current and instantaneous over current relays) Microprocessor (block diagram and flow chart of overcurrent relay) – Fundamentals of Numerical relay Principles of overcurrent, directional, distance and differential Circuit breakers – operating principle – arc phenomenon – arc extinction – principle & methods Circuit breaker classification based on medium of arc extinction – SF6 & VCB	**11**
4	Load flow studies – Introduction- Types of buses - Network model - admittance matrix- Gauss Siedal method of load flow analysis (Qualitative analysis)– (numerical problems not required) Power system stability - steady state, dynamic and transient stability power angle curve - steady state stability limit - swing equation Equal area criterion and application - methods of improving stability limits	**11**

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Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

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Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Learn different types of power generating systems and schedule generation appropriate for a given area.	**K2**
CO2	Understand the electrical performance of any transmission line.	**K2**
CO3	Demonstrate the working of switchgear for protection schemes.	**K2**
CO4	Analyse the voltage profile of any given power system network using iterative methods.	**K3**
CO5	Analyse the steady state and transient stability of power system networks.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	**3**					**2**	**1**	**2**			**1**	**1**
CO2	**3**	**3**										**1**
CO3	**3**	**1**				**2**		**2**				**1**
CO4	**3**	**3**	**2**		**1**							**1**
CO5	**3**	**3**	**2**		**1**							**1**

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

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Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Electrical Power Systems	Wadhwa C. L.	New Age International	8th edition 2023
2	Principles of Power System	V. K. Mehta and Rohit Mehta	S.Chand	4th edition reprint 2020
3	Power System Protection and Switchgear	Badri Ram and D. N. Viswakarma	Tata McGraw Hill	2nd edition, 2011
4	Non-conventional energy sources	B. H. Khan	Tata McGraw Hill	3rd edition, 2017
5	Power System Analysis	Hadi Saadat	McGraw Hill	2nd edition, 2002.
6	Modern Power System Analysis	D. P. Kothari and I. J. Nagrath	McGraw Hill	2nd edition, 2002.
7	Power System Analysis and Design	Gupta B. R.,	S. Chand	2006

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